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Personalized Federated Training of Diffusion Models with Privacy Guarantees

Kumar Kshitij Patel[1,*] Bingqing Jiang[2,*] A F M Mahfuzul Kabir[3,*] Weitong Zhang[4]
Difan Zou[2] Lingxiao Wang[3]
[1]Yale FDS [2]The University of Hong Kong [3]New Jersey Institute of Technology
[4]University of North Carolina at Chapel Hill
kkpatel@ttic.edu bingqingjiang@connect.hku.hk dzou@hku.hk
{ak3535, lw324}@njit.edu weitongz@unc.edu

Abstract

We propose a federated framework for training diffusion models on decentralized and private datasets. The method learns a shared generative model alongside personalized client models, allowing clients to benefit from cross-client structure while ensuring that the shared model cannot reproduce any client's data on its own. We provide formal differential privacy guarantees for each client and establish utility bounds for conditional generation under a Gaussian mixture model, showing that collaboration improves sample quality relative to private non-collaborative training. Experiments on CIFAR-10, Colorized MNIST, and CelebA support these results: the method generates high-fidelity samples, improves performance on minority and underrepresented classes, and maintains strong protection against membership inference, memorization, and reconstruction attacks.

1. Introduction

Modern machine learning systems rely on large and diverse datasets, and this reliance has driven the development of foundation models across many domains [8, 46, 47]. In practice, however, institutions often cannot pool data in a central location. Hospitals, financial organizations, and scientific laboratories operate under strict legal and regulatory limits, including privacy laws, copyright restrictions, compensation requirements [9, 28, 33, 99]. These constraints have reduced broad public data availability [94]. At the same time, many fields face intrinsic scarcity, as rare diseases, small cohorts, and costly experiments limit the amount of data any institution can collect [17, 19, 75, 77]. Consequently, models trained on fragmented datasets often struggle to capture rare or underrepresented cases.

Federated learning (FL) helps institutions train machine learning models without sharing raw data [43, 64, 65]. And further combining FL with Differentially private training reduces what a learned model reveals about individual clients. FL works well when institutions need only a small number of downstream predictors; however, many real applications require a broad suite of tasks, including risk prediction, anomaly detection, subgroup analysis, and related objectives that rely on the same underlying data. Training a new federated model for each task is expensive and repeatedly consumes privacy budget, since each model release introduces additional privacy loss. This practical limitation motivates a different strategy: rather than solving each downstream task collaboratively, can institutions train a generative model that expands data coverage and supports multiple downstream tasks without further collaboration?

This direction places synthetic data generation at the center. A shared generative model, if trained with formal privacy guarantees, could create data that supplements rare modes, strengthens underrepresented regions, and remains safe to share across institutions. Diffusion models [39, 85] form a strong basis for this goal, and recent work has explored their training in federated or privacy-constrained settings [22, 55, 81, 82, 95, 97]. These studies demonstrate empirical progress, yet they train a single global diffusion model that provides no client-specific control. Even worse, diffusion models can memorize sensitive training examples [10, 18, 72], so releasing a fully shared generator can expose all clients to extraction or reconstruction attacks unless the entire generative pipeline satisfies strong privacy guarantees. Standard DP training techniques for diffusion models [26, 67, 89, 97] do not provide such protection: they often degrade sample quality, scale poorly to high-dimensional data, and still leave models vulnerable to memorization. These issues make a single global generator both risky and inflexible in decentralized environments.

This paper proposes a federated framework for train-

*Equal contribution.

ing diffusion models that addresses these limitations. The method supports decentralized training with formal privacy guarantees, yielding a personalized generative model for each client, as well as a shared model that facilitates collaboration. The shared model remains private and cannot generate samples from any client's distribution on its own; it becomes useful only when a client pairs it with its local model. This design enhances safety, strengthens client-specific control, and enables the generation of synthetic data without repeated losses of the privacy budget, allowing institutions to support a broader range of downstream tasks more efficiently. Our main contributions are as follows:

1. **Personalized federated training of diffusion models.** We introduce a federated approach for training diffusion models on decentralized and private datasets. The method personalizes the reverse diffusion process (i.e., de-noising) by splitting it into client-specific and shared components: each client model maps noisy client images to clean ones, while the shared model maps standard Gaussian noise to a mixture of noisy client images. As a result, the shared model only ever processes noisy client images, thereby reducing the risk of memorization and preventing it from generating client-specific samples on its own. This structure provides clients with direct control over their synthetic data, while allowing the shared model to capture higher-level and generalizable cross-client features that help mitigate dataset imbalance.
2. **Formal privacy and utility guarantees.** We show that our method provides differential privacy for each client's data, with the privacy budget determined by the diffusion parameters and the degree of personalization (Theorem 5.1). We also establish utility guarantees for learning Gaussian mixture models (Theorem 5.2), showing that the utility interpolates between non-collaborative training (the private extreme) and centralized training (completely non-private). Specifically, we demonstrate that our method enhances generative quality compared to non-collaborative baselines by reducing the quadratic 2-Wasserstein distance to the actual data distribution.
3. **Empirical validation of utility and privacy.** Experiments on CIFAR-10, Colorized MNIST, and CelebA show that our method generates high-quality images, outperforming non-collaborative training across varying data sizes and levels of heterogeneity. Privacy evaluations—including membership inference, memorization, and reconstruction attacks—confirm that the design protects against known diffusion-model vulnerabilities.

Notation. For $x \in \mathbb{R}^d$, $\|x\|_2$ denotes its ℓ_2 norm, and $x[i]$ denotes its i-th coordinate. $[M]$ denotes the set $\{1, \dots, M\}$, and $\mathcal{P}, \mathcal{B}$ denote index sets. $\mathcal{U}(1, T)$ denotes the uniform distribution over $[T]$.

2. Related Work

Image Generation and Diffusion Models. Image generation has long served as a core challenge in machine learning, motivating the development of autoregressive models [90, 91], variational autoencoders [52, 92], and generative adversarial networks [32]. Diffusion models and score-based methods have recently become the dominant approach because they offer stable training, a principled probabilistic formulation, and state-of-the-art sample quality [24]. This work focuses on denoising diffusion probabilistic models (DDPMs) [39], which form the foundation for many modern diffusion architectures. Although we illustrate our method using DDPMs, the underlying ideas apply broadly across diffusion-based generative models.

Federated Learning of Diffusion Models. Federated learning (FL) enables collaborative training on decentralized datasets while avoiding the exchange of raw data [43, 64, 65], a design motivated in part by data regulations such as GDPR [28]. While early FL research focused on predictive models, several recent efforts have applied FL to diffusion models. de Goede et al. [22] extended FedAvg to train DDPMs and showed that a global model can approach the quality of centralized training while reducing communication costs. Vora et al. [95] proposed FedDM, which incorporates quantization and proximal regularization to improve communication efficiency and robustness under heterogeneous data, and Lai et al. [55] refined quantization strategies for cross-device FL. However, these methods—and FL protocols more broadly—still leave clients vulnerable to gradient-based reconstruction attacks [100], since they do not provide formal differential privacy guarantees. To address this, Sattarov et al. [81, 82] incorporated differentially private SGD [1] into FedAvg for diffusion models on tabular data. This provides differential privacy, but still relies on a single global model, and extending DP-SGD–based diffusion training from low-dimensional tabular settings to high-dimensional image generation is non-trivial, because DP noise can destabilize the denoising process. Altogether, these approaches advance federated diffusion training, but important challenges remain: end-to-end generation through a global diffusion model still carries significant memorization risk, as demonstrated by extraction attacks [10], and relying on a single generator precludes personalized control over the synthetic data clients produce.

Personalization and Differential Privacy. Personalization is a widely used strategy in FL to address data heterogeneity and provide client-specific control [69]. Most personalized FL frameworks partition the model into shared components trained collaboratively and local components tailored to each client. Personalization can also improve the

privacy–utility trade-off, since an appropriately designed shared model can focus on generalizable structure while leaving fine-grained, client-specific features to the local model [7]. Jothiraj and Mashhadi [42] applied this idea to diffusion models by splitting the UNet into shared and local modules, and Chen et al. [13] proposed a more sophisticated scheme for one-shot personalization in federated settings. Allmendinger et al. [3] developed a collaborative training strategy that distributes parts of the diffusion training pipeline across clients and server to reduce computational overhead. However, none of these approaches provides formal differential privacy guarantees. In the centralized setting, several works (e.g., [41, 60]) develop differentially private training methods for DMs. Nevertheless, they cannot be directly applied in the FL setting and instead rely on public datasets for private fine-tuning, making direct comparisons less fair [88].

We discuss other less related works in Appendix A.

3. Preliminaries

Denoising Diffusion Models. Denoising diffusion models aim to learn a distribution p_θ that approximates a target distribution q_0. The framework consists of a forward diffusion process and a reverse denoising process. In the forward process, a sample $x_0 \sim q_0$ is gradually corrupted by Gaussian noise to produce latent variables $\{x_t\}_{t=1}^T$ according to

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, z_t \ , \qquad z_t \sim \mathcal{N}(0, I) \ , \quad (1)$$

where $\bar{\alpha}_t$ is the cumulative product of a noise schedule $\{\beta_t\}_{t=1}^T$ (see Appendix B). For sufficiently large T, and an appropriate noise schedule, the distribution of x_T approaches a standard Gaussian. The reverse process then defines a parameterized Markov chain[1]

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta(x_t, t),\, \Sigma_\theta(x_t, t)) \ , \quad (2)$$

and learning aims to approximate these reverse transitions.

Following Ho et al. [39], one may set $\Sigma_\theta(x_t, t) = (1 - \bar{\alpha}_t)I$, which reduces the learning problem to training a neural denoiser z_θ that predicts the Gaussian noise z. Concretely, we train z_θ via

$$\mathbb{E}_{t\sim\mathcal{U}(1,T),\, x_0,\, z_t}\left[\|z_t - z_\theta(x_t, t)\|_2^2\right] \ , \quad (3)$$

and then express the mean $\mu_\theta(x_t, t)$ of the reverse conditional in terms of x_t, $z_\theta(x_t, t)$, and the known coefficients of the diffusion process. Given a trained z_θ, one can draw new samples by iteratively sampling from the reverse conditionals $p_\theta(x_{t-1} \mid x_t)$ starting from $x_T \sim \mathcal{N}(0, I)$.

[1] For a broader introduction to DDPMs and justification of this reverse sampling formulation, we refer the reader to existing surveys [49, 71].

Problem Setup. We consider a federated learning scenario with M clients. Each client $m \in [M]$ holds a private dataset D_m, with samples drawn independently from an underlying distribution q_0^m. The goal is to learn a collection of generative models $(p^m)_{m\in[M]}$ that approximate $(q_0^m)_{m\in[M]}$ while preserving the privacy of each client's data. Clients collaborate through a shared server, but no client reveals raw samples. Since each client has a distinct data distribution, in general both the individual samples within each D_m and the distribution-specific characteristics that distinguish one client's data from another need to be protected. Note that our paper mainly focus on the cross-silo setting [44].

Differential Privacy. To formalize privacy in the federated multi-distribution setting, we use two notions: central differential privacy (central DP) [27] and local differential privacy (local DP) [50]. The definitions are as follows.

Definition 3.1 ((ϵ, δ)-DP)**.** *A randomized mechanism $\mathcal{A}$ satisfies (ϵ, δ)-differential privacy if, for any two adjacent datasets D and D' that differ in one element, and any measurable set O: $\mathbb{P}[\mathcal{A}(D) \in O] \leq e^\epsilon\, \mathbb{P}[\mathcal{A}(D') \in O] + \delta$.*

In federated learning, people often consider sample-level central DP [31, 62, 70, 98], which aims to protect each individual data point and typically requires a trusted server. Others consider client-level central DP [43, 66], which aims to protect the entire dataset of each client. Although client-level central DP corresponds to a stronger privacy target, as its protected unit is larger, it still relies on a trusted server and usually leads to worse utilities. In our paper, we aim to provide the following Local DP guarantee, which does not require a trusted server.

Definition 3.2 ((ϵ, δ)-LDP)**.** *A randomized mechanism $\mathcal{A}$ satisfies (ϵ, δ)-local differential privacy if, for any two inputs x and x', and any measurable set O: $\mathbb{P}[\mathcal{A}(x) \in O] \leq e^\epsilon\, \mathbb{P}[\mathcal{A}(x') \in O] + \delta$.*

Local DP requires each client to randomize its data before any communication occurs, ensuring privacy for individual data points *prior* to any aggregation. In the multi-distribution setting, this means the server never receives information that can reliably distinguish between any two possible inputs from a client. Thus, Local DP does not rely on a trusted server (as opposed to client-level central DP). Moreover, per-sample Local DP immediately implies the same level of sample-level central DP (making it strictly stronger). MOtivated by this, we want to provide sample-level DP since it is the most useful for cross-silo setting, where each silo/client (e.g., hospital) has data records from many different individuals (e.g., patients).

4. Proposed Method

Our method, PFDM (Algorithm 1), consists of two stages.

Stage 1: Training client-specific denoisers. Each client $m \in [M]$ trains a personalized denoiser z_{θ_m} on its private dataset D_m using the standard DDPM training routine T-DDPM (Algorithm 3 in Appendix B). This model remains local and is never shared. After training, client m constructs a noisy dataset $\tilde{D}_m$ by first applying clipping [1] to bound data magnitude and then running the forward diffusion process for t_0 steps (lines 3–8 in Algorithm 1). This procedure produces samples drawn from the diffused distribution

$$q_m(x_{t_0}) := \int q_0^m(x_0) q(x_{t_0}|x_0) dx_0 \ , \tag{4}$$

where q_0^m is the input data distribution for client m, and

$$q(x_t|x_0) := \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I) \ . \tag{5}$$

This step ensures that the injected noise in the dataset is calibrated to satisfy the differential privacy requirement. Each client then sends only $\tilde{D}_m$ to the server.

Stage 2: Training a shared global denoiser. The server aggregates the noisy datasets $\{\tilde{D}_m\}_{m\in[M]}$ and trains a global denoiser z_w (lines 11–14 in Algorithm 1). Because each $\tilde{D}_m$ is differentially private, the learned z_w is also differentially private and can be safely shared with all clients.

Note that during the training, our method only requires one round of communication at the beginning.

Sampling with PFDM. To generate synthetic data for client m, PFDM first uses the global denoiser z_w to run the reverse diffusion process for T steps, producing an intermediate sample $\tilde{x}_0$ (Algorithm 2). Client m then applies its personalized denoiser z_{θ_m} for t_0 steps to obtain a final sample that reflects both the collaborative global structure and the fine-grained details specific to client m. In settings where z_{θ_m} can already generate high-quality samples from noise, clients may rely solely on their local model.

Why does the split work? The diffused distributions $\{q_m(x_{t_0})\}$ preserve coarse, high-level visual structure while discarding more fine-grained features, because during the diffusion process, the image's fine-grained details (e.g., textural details) are perturbed faster than the macro structures such as the background [78]. This effect causes the diffused distributions to concentrate on similar large-scale features even when the original client datasets differ substantially. Training z_w on these privatized diffused datasets therefore allows the shared model to capture structural patterns that are broadly useful without accessing sensitive information. Each client then applies its personalized denoiser z_{θ_m} to recover the sensitive fine-scale details specific to its own distribution. We further discuss this coarse-to-fine hierarchy of diffusion denoising, which has been observed and exploited in many domains, in Appendix C.

The privacy–utility trade-off. The hyperparameter t_0 determines how much information survives the diffusion step:

Algorithm 1 **P**ersonalized **F**ederated Training of **D**iffusion **M**odel (PFDM)

input Training dataset $\{D_m\}_{m\in[M]}$, shared model parameter w, client-specific model parameters $\{\theta_m\}_{m=1}^M$, local time step t_0, global time step T, noise schedule $\{\beta_t\}_{t=1}^T$, clipping parameter C

1: **on client** $m \in [M]$ **do**
2: Train a personalized secret denoiser z_{θ_m} = T-DDPM$(D_m, T, \{\beta_t\}_{t=1}^T, \theta_m)$
3: Sample n_m data from D_m indexed by $\mathcal{B}_m$ and set $\bar{\alpha}_t = \prod_{s=1}^t (1-\beta_s)$
4: **for** $i \in \mathcal{B}_m$ **do**
5: $\tilde{x}_0^{i,m} = \sqrt{\bar{\alpha}_{t_0}} \cdot \text{CLIP}(x_0^{i,m}, C) + \sqrt{1-\bar{\alpha}_{t_0}} z$, where $z \sim \mathcal{N}(0, I)$, $\text{CLIP}(x, C) = x \cdot \min\{1, C/\|x\|_2\}$
6: **end for**
7: **Send (communicate)** $\tilde{D}_m = \{\tilde{x}_0^{i,m}\}_{i\in\mathcal{B}_m}$ to server
8: **end on client**
9: **on server do**
10: Obtain $\tilde{D} = \{\tilde{D}_m\}_{m\in[M]}$ and train z_w = T-DDPM$(\tilde{D}, T, \{\beta_t\}_{t=1}^T, w)$
11: **end on server**

output Shared global denoiser z_w

Algorithm 2 Sampling Procedure for Personalized Federated Diffusion Model

input Shared global denoiser z_w, personalized denoiser z_{θ_m}, global time step T, local time step t_0 noise scheduling parameters $\{\beta_t, \sigma_t\}_{t=1}^T$

1: $\tilde{x}_0 =$ S-DDPM$(z_w, T, \{\beta_t, \sigma_t\}_{t=1}^T)$
2: Set $\bar{\alpha}_t = \prod_{s=1}^t (1-\beta_s)$ and let $x_{t_0} = \tilde{x}_0$
3: **for** $t = t_0, \ldots, 1$ **do**
4: $x_{t-1} = \frac{1}{\alpha_t}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} z_{\theta_m}(x_t, t)\right) + \sigma_t z$, where $z \sim \mathcal{N}(0, I)$ if $t > 1$ else $z = 0$
5: **end for**

output x_0

larger values increase the injected noise and strengthen privacy, whereas smaller values preserve more detail. We will next formalize this privacy–utility trade-off.

5. Main Results

5.1. Privacy Guarantee

We first state the privacy guarantee of our method.

Theorem 5.1 (Privacy Guarantee of PFDM)**.** *Given the training datasets $D = \{D_m\}_{m\in[M]}$, if we choose the local diffusion depth t_0, total number of steps T, noise schedule $\{\beta_t\}_{t=1}^T$, and clipping parameter C, then, for all $x \in D$, the output of Algorithm 1 satisfies*

$$\left(\frac{2C^2}{\sigma^2} + C\sqrt{\frac{8\log(1/\delta)}{\sigma^2}}, \ \delta\right)\text{-LDP} \ , \tag{6}$$

where $\sigma^2 = (1-\bar{\alpha}_{t_0})/\bar{\alpha}_{t_0}$ *and* $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

The guarantee in Theorem 5.1 provides an explicit upper bound on the privacy parameter ϵ. The parameter $\sigma^2 = (1-\bar{\alpha}_{t_0})/\bar{\alpha}_{t_0}$ is the variance of the effective additive noise applied to each datum after t_0 steps of the forward diffusion process. A larger σ^2 yields stronger privacy, and achieving this requires a longer diffusion process (larger t_0) before sending the noisy data to the server. Thus, t_0 acts as a knob for the privacy–utility trade-off (also see utility guarantee in the next sub-section).

For example, with $T = 1000$, a linear noise schedule, and clipping parameter $C = 10$, choosing $t_0 = 690$ yields an (ϵ, δ)-LDP guarantee with $\epsilon = 10$ and $\delta = 10^{-5}$. Although $\epsilon = 10$ seems large in the context of central DP, it represents a *local* DP guarantee on every data point, which is significantly stronger. In practical scenarios, values of ϵ in the tens or hundreds can still effectively defend against strong membership inference attacks [40, 63]. We also provide empirical evaluations demonstrating the robustness of our method against various privacy attacks.

5.2. Utility Guarantee

To characterize the utility of the learned diffusion models under privacy constraints, we follow a line of recent work that theoretically analyzes diffusion models through Gaussian mixture models (GMMs) and their variants [14, 16, 29, 35, 37]. We assume each client $m \in [M]$ aims to sample from a mixture distribution $q = \sum_{y\in\mathcal{Y}} w_y \mathcal{N}(\mu_y, I_d)$, where $\mathcal{Y} = \{1, \dots, K\}$, $\sum_y w_y = 1$, and each class $y = k$ corresponds to the Gaussian component $\mathcal{N}(\mu_k, I_d)$. We consider the conditional generation setting in which each client wishes to perform well on all classes $k \in [K]$, including those that are underrepresented locally. Let n_k^m denote the number of class-k samples on client m, and let $N_k = \sum_m n_k^m$ be the total number of class-k samples across all clients. The goal of PFDM is to train a global denoiser z_w and private client-specific denoisers $\{z_{\theta_m}\}$ that together learn conditional distributions $p^m(\cdot \mid y = k)$ approximating $q(\cdot \mid y = k)$ for all $m \in [M]$.

Following Chen et al. [12], we analyze a piecewise linear parameterization of the denoising network,

$$\epsilon_\theta(x, y) = \sum_{k=1}^{K} \mathbf{1}_{y=k}\big[W^k x + V^k c(y)\big] \ , \qquad (7)$$

where $c(y)$ is the one-hot encoding of the label and $\theta = \{W^k, V^k\}_{k=1}^K$. Utility is measured via the 2-Wasserstein distance W_2^2, which quantifies sampling error between the true and generated distributions [12, 38].

Theorem 5.2 (Utility Guarantee of PFDM on GMMs)**.** *For any class $k \in \mathcal{Y}$ and sufficiently large T, suppose each client m has $n_k^m = \Omega(d\log(1/\zeta))$ samples of class k, each with bounded ℓ_2-norm C. Under the same choices of t_0, $\{\beta_t\}$, and C as in Theorem 5.1, let $p^m_{\tilde{D},D_m}$ denote the distribution learned for client m. Then*

$$\mathbb{E}_{\tilde{D},D_m}\Big[W_2^2\big(q;\, p^m_{\tilde{D},D_m}\big) \mid y = k\Big] = O\bigg(\frac{2}{2+3\sigma^2}\cdot\frac{d^2}{N_k} \ + \ \frac{3\sigma^2}{2+3\sigma^2}\cdot\frac{d^2}{n_k^m}\bigg), \qquad (8)$$

with probability at least $1-\zeta$, where $\sigma^2 = (1-\bar{\alpha}_{t_0})/\bar{\alpha}_{t_0}$.

Theorem 5.2 shows that the utility of PFDM is governed by the noise level σ^2, and in turn by the diffusion depth t_0:

- As $\sigma^2 \to \infty$ (maximal privacy), the bound approaches the *non-collaborative* rate $O(d^2/n_k^m)$.
- As $\sigma^2 \to 0$ (minimal privacy), the bound approaches the *centralized* rate $O(d^2/N_k)$.
- For intermediate σ^2, the guarantee interpolates smoothly between these two extremes.

Because N_k can be much larger than n_k^m—especially when class k is underrepresented on client m—collaboration provides substantial gains for minority classes. In particular, comparing with non-collaborative training, let $p^m_{D_m}$ denote the model trained solely on client m's data. Under the same conditions and, additionally, assuming $N_k = \Omega((1+\sigma^2)n_k^m)$ and $n_k^m = \Omega\big((1+\sigma^2)^2 d\log^{3/2}(1/\delta)\big)$, we obtain

$$\mathbb{E}_{\tilde{D},D_m}\Big[W_2^2\big(q;p^m_{D_m}\big) - W_2^2\big(q;p^m_{\tilde{D},D_m}\big) \mid y = k\Big] = \Omega\bigg(\frac{\sigma^2+1}{(2\sigma^2+1)^2}\cdot\frac{d^2}{n_k^m}\bigg), \qquad (9)$$

showing that PFDM outperforms non-collaborative learning whenever class k has sufficient overall support.

6. Experiments

We evaluate PFDM on CIFAR-10 [54], Colorized MNIST [4], and CelebA [61], measuring both the quality of the generated samples and the resilience of the method to privacy attacks.

6.1. Utility Evaluations

Data Preparation. To introduce heterogeneous data among clients, we partition each dataset into two clusters and assign each client a majority of samples from one cluster and a minority from the other.

- *CIFAR-10:* Cluster 1 contains {airplane, car, ship, truck} and Cluster 2 contains {bird, cat, deer, dog, frog, horse}. We create four clients: two receive 1,250 samples per class from Cluster 1 and 125 per class from Cluster 2; the other two use the reverse allocation.
- *Colorized MNIST:* Digits are split into $\{0, 1, 6, 7, 9\}$ and $\{2, 3, 4, 5, 8\}$. We create two clients. One has both

Table 1. FID scores for different methods on different datasets (lower is better).

	CIFAR-10	Colorized MNIST	CelebA
METHODS	FID (Major/Minor/Avg)	FID (Major/Minor/Avg)	FID (Major/Minor/Avg)
Non-private (Centralized)	16.27/17.62/16.95	1.85/1.45/1.66	13.72/11.70/12.71
Non-private (FedDM)	18.05/19.15/18.60	1.89/1.51/1.70	14.47/11.83/13.15
Non-collaborative	19.87/36.44/28.16	2.19/5.99/4.09	23.42/41.38/32.40
Ours	19.85/35.78/27.82	1.72/4.79/3.26	18.11/28.09/23.10

red and blue digits from $\{0, 1, 6, 7, 9\}$ with 1,000 samples per digit–color combination (majority), plus blue digits from $\{2, 3, 4, 5, 8\}$ with 10 samples per combination (minority). The second client mirrors this pattern but with $\{2, 3, 4, 5, 8\}$ as majority and blue digits from $\{0, 1, 6, 7, 9\}$ as minority.

- *CelebA:* We split by gender and hair color and form two clients. The first receives 5,000 male–black-hair, 5,000 male–non-black-hair, and 50 female–non-black-hair images, with female–black-hair unseen. The second receives 5,000 female–black-hair, 5,000 female–non-black-hair, and 50 male–non-black-hair images, again producing clear majority/minority groups.

Conditional Generation. We evaluate conditional image generation for all classes. Since we focus on protecting image content rather than labels, labels are treated as public and used as conditioning variables when training the diffusion models. Architectural details appear in Appendix D.

Baselines. We compare the following:

1. **Centralized Baseline (Non-Private Extreme):** A single model trained on the union of all clients' datasets.
2. **Federated Baseline (Non-Private):** A single model trained using the federated training method [95].
3. **Non-Collaborative Baseline (Private Extreme):** Independent models trained on each client's data.
4. **Collaborative (Ours):** Collaboration without sharing raw data. This includes (i) a global model trained on noisy, clipped client data (providing DP guarantees) and (ii) a personalized local model trained on each client's clean data. Because the global model sees only clipped images, its outputs tend to remain clipped; we therefore augment local training with a mixture of clipped and unclipped samples and an auxiliary conditioning signal that steers generation toward unclipped images.

Training. All methods use a linear noise schedule with $T = 1000$. We fix the privacy budget at $\epsilon = 10, \delta = 10^{-5}$. More details can be found in Appendix D.

Evaluation Metrics. We use the Fréchet Inception Distance (FID) [38, 39] to evaluate image quality. We report class-conditional FID for the majority and minority groups of the first client, using 1,000 real images per class. We also consider the downstream classification tasks to evaluate the utilities, and we report the results in Appendix D.

Results. Figure 1 shows CelebA samples. Our collaborative method produces high-quality images for both majority and minority groups, noticeably surpassing the non-collaborative baseline on minority classes. Additional qualitative samples for CIFAR-10 and Colorized MNIST appear in Appendix D. Table 1 summarizes FID results across datasets. Our method achieves scores close to the non-private baselines and substantially better than the non-collaborative baseline, especially for minority groups.

Effect of the Number of Clients. We study how performance changes as the number of clients increases while keeping the total dataset size fixed. Using CIFAR-10, we vary the M from 4 to 8, 16, 32, 64, and 128. As shown in Figure 2, the performance gap between our collaborative method and the non-collaborative baseline widens as the number of clients increases, demonstrating the growing value of collaboration under extreme data heterogeneity.

Comparison with DP Training Baselines. We additionally compare our approach against a differentially private diffusion training baseline, DPDM [26]. In particular, we implement DPDM in a federated setting by training the diffusion model collaboratively using standard private federated learning algorithms, i.e., private variant of Minibatch SGD [48, 73]. Using MNIST (with the same two-client structure as Colorized MNIST but without colorization), our method achieves FID scores of 5.40 and 8.51 on the majority and minority groups (average 6.96) under $\epsilon = 10$. In contrast, the DPDM-based federated baseline attains 31.06 and 36.40 (average 33.73), showing that direct DP training of diffusion models in FL yields substantially worse image quality than our collaborative framework. We report the results of the centralized private baseline in Appendix D.

6.2. Privacy Evaluations

Global Output. Figure 3 shows samples generated by the global model on Colorized MNIST (see Figure 6 in Appendix D for additional examples). The images preserve only coarse color and layout, while the digit shapes remain unrecognizable. This behavior shows that the global model does not reveal identifiable information and provides strong



Figure 1. CelebA samples generated by different methods. (a), (b) correspond to samples generated for majority class group: Black Hair and Male. (c), (d) correspond to samples generated for minority class group: NonBlack Hair and Female. We report the results for the model trained by the first client.

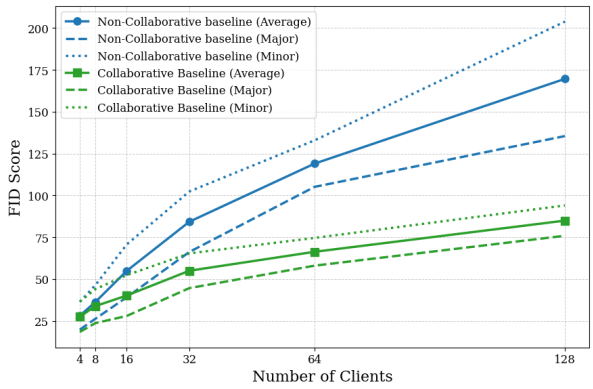


Figure 2. FID scores under different number of clients. The results show an increased difference between the Non-Collaborative and Collaborative baselines with an increasing number of clients.

protection for client data.

Membership Inference Attack. Membership inference attacks (MIA) attempt to determine whether a specific sample appeared in a model's training set and therefore pose a direct threat to data privacy. In diffusion models, such attacks exploit the tendency of the model to fit training samples more closely than unseen data, which yields lower reconstruction error or smaller deviations along the denoising trajectory. We adopt the Proximal Initialization Attack (PIA) [53], a query-efficient and highly discriminative MIA method for diffusion models (see Appendix D for details). For evaluation, we create a balanced query set containing equal numbers of training and held-out samples.

Table 2 reports the results of PIA on our method using three standard metrics: Area Under the ROC Curve (AUC), Attack Success Rate (ASR), and the True Positive Rate at 1% False Positive Rate (TPR@1% FPR). Across all datasets, the attack performance remains close to random guessing (50%), which indicates minimal privacy leakage and shows that our framework provides strong resistance to MIA. For comparison, Table 6 in Appendix D shows the vulnerability of standard nonprivate models. As training progresses, privacy leakage grows substantially. On CIFAR-10, the AUC rises from 53.03% to 82.13% after 1000 epochs, and on Colorized MNIST and CelebA the AUC reaches 99.62% and 99.59%, respectively. These results highlight the significant privacy risk of nonprivate diffusion training and the robustness of our method.

Memorization. Memorization refers to a generative model's tendency to produce outputs that closely reproduce specific training samples. To detect potential memorization, we use the nearest-neighbor ratio criterion proposed in [102], which correlates well with human visual judgments (see Appendix D for details). For each dataset, we generate twice as many samples as were used during training and evaluate them under this criterion. Across all datasets, none of the generated samples satisfy the memorization condition, indicating that **our method avoids direct replication of training examples**.

Reconstruction Attack. Reconstruction attacks aim to recover the original input data from perturbed or intermedi-

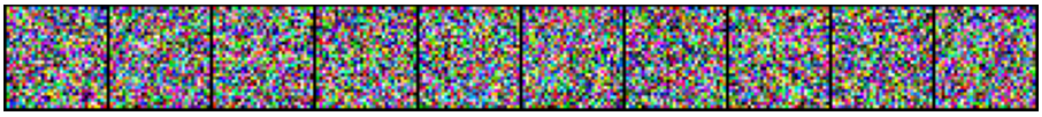


Figure 3. Samples generated by the global model used in our framework with red digits 0 to 9. The output digit shapes remain unrecognizable, indicating strong privacy protection of our method.

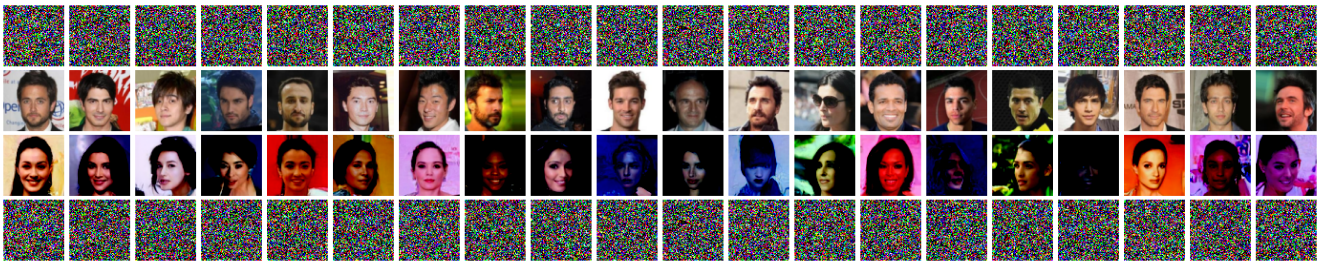


Figure 4. Reconstruction results on CelebA. The four rows show: (1) the noisy images uploaded by the client, (2) the original clean images, (3) reconstructions produced by the pretrained-model attacker, and (4) reconstructions produced by the global-model attacker.

Table 2. MIA results for our method (global model used in our framework trained for 300 epochs). 50% AUC/ASR means the random guess. Lower TPR@1% FPR indicates stronger privacy protection. MIA on a standard nonprivate model (centralized method) achieves 82.13%,99.62%,99.59% AUC for CIFAR-10, Colorized MNIST, and CelebA (see Table 6 in Appendix D).

	CIFAR-10	Colorized MNIST	CelebA
AUC	50.01	49.70	50.08
ASR	50.15	50.10	50.34
TPR@1% FPR	0.82	1.07	0.86

Table 3. Recovery score for various reconstruction attacks on the noisy image received by the server: "None" (no attack), "Global" (attack using the collaboratively trained global model in our framework), and "Pretrained" (attack using a denoiser pretrained on a similar dataset). Lower recovery score indicate better reconstructions (0 means perfect reconstruction). Reconstruction examples appear in Figure 4, also see Figures 8, 9 in Appendix D.

	CIFAR-10	Colorized MNIST	CelebA
METHODS	Score	Score	Score
None	2.43	3.77	1.13
Global	2.41	3.66	1.12
Pretrained	1.39	1.20	0.89

ate representations, such as noisy samples exchanged during collaborative training. These attacks pose a significant privacy threat in federated settings, where raw data never leaves the client but may still be inferred from shared information. In our evaluation, we simulate a server-side adversary that attempts to reconstruct client images from the noisy data uploaded during the diffusion process.

We study two adversarial models: (1) a *global model attacker* that uses the diffusion model trained collaboratively across all clients, representing an adversary with full access to the DP model parameters; and (2) a *pretrained model attacker* that uses a denoiser trained on a related dataset, representing an adversary with partial knowledge of the data distribution. To quantify reconstruction quality, we use the recovery score, defined as the average normalized ℓ_2 distance between each original sample and its reconstruction: $n^{-1}\sum_{i=1}^{n}\|x_0^i - x_{\text{rec}}^i\|_2/\|x_0^i\|_2$.

Figure 4 presents reconstruction outputs for CelebA. Our method prevents both adversaries from producing recognizable reconstructions, including the stronger pretrained-model attacker. The recovery scores in Table 3 support this observation and show that our framework maintains strong privacy protection. See additional results in Appendix D.

7. Conclusion and Future Work

We introduced a framework for collaboratively training diffusion models across decentralized and private datasets. Our method combines personalization with a privacy-preserving global model, providing formal privacy and utility guarantees. Experiments across multiple datasets demonstrate that the framework generates high-quality images, improves performance for minority groups, and provides strong protection against privacy attacks. Our work leaves several open directions, like extending the utility analysis to broader model classes, further strengthening the privacy guarantees, scaling to high-resolution datasets, and accounting for strategic behavior [36].

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